

Chapter - 6

Time value of money.

* Amortisation $\rightarrow$ Sect C 10m

* Compound Interest $\rightarrow$ Sect B 7m.

XYZ Co. borrows £ 10 000 000 at the rate of interest 15%. The loan is to be repaid 5 equal installment. Paid at the end of each year. Calculate loan amortisation

Given:

Loan Borrowed = 10 000 000

Interest = 15%

Amnt per installment = ?

Amortisation table = ?

Annuity.

Present value Annuity = $PMT \times PVIFA\% n.$

$$10 000 000 = PMT \times 3.352$$

$$Payment\ time = \frac{10 000 000}{3.352}$$

$$= 298 329$$

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Year	Beginning Amnt (i)	Annual Installment	Interest 3 = 15%1	Present value of Payment 4 = 2-3	Beginning of Borrowed Capital 5 = 1-4
1	10 000 000	298 329	1 500 000	1 483 29	8 516 71
2	8 516 71	298 329	1 277 51	1 705 78	6 810 93
3	6 810 93	298 329	1 021 64	1 961 65	4 849 28
4	4 849 28	298 329	727 39	2 255 90	2 593 38
5	2 593 38	298 329	389 01	2 594 28	- 90

2. A Go name Nimal took a loan for purchasing a van costing ₹ 10,00,000 at 14% interest rate. The loan is to be repaid over next 5 years. Estimate the instalment and prepare loan amortization schedule.

Present value Annuity = $PMT \times PVIFA_{14\%, 5}$

$10,00,000 = PMT \times 3.433$

Payment time = $\frac{10,00,000}{3.433}$

= 291290

Year	Beginning (1) Amt	Annual Instalment (2)	Interest 3 = (1) × 14%	Per Capita Payment 4 = 2-3	Beginning & Ending Capital 5 = 1-4
1	10,00,000	291290	1,40,000	1,51,290	8,48,710
2	8,48,710	291290	1,18,819	1,72,471	6,76,239
3	6,76,239	291290	94,674	1,96,616	4,79,623
4	4,79,623	291290	67,147	2,24,443	2,55,180
5	2,55,180	291290	35,767	2,55,522	-43

3. Mr. Homeprakash borrowed ₹ 5,00,000 at an interest rate 14% p.a. to purchase a car. The bank provides in 6 annual installment facilities to repay the loan amount. You are required to calculate loan amortization and 1) Int amt 2) amount repayable to bank at the end of 6 years.

Factors determine.

Present value Annuity = $PMT \times PVIFA \% n.$

$500000 = PMT \times 3.889$

$PMT = \frac{500000}{3.889}$

$= 128568$

Year	Beginning Amt	Annual Installment	Interest	Per value Payment	Beginning of Borrowed Capital
1	500000	128568	3 = 14% x 1	4 = 2-3	
2	441432	128568	70000	58568	441432
3	374664	128568	61800	66768	374664
4	298549	128568	52453	76115	298549
5	211778	128568	41797	86771	211778
6	112859	128568	29649	98919	112859
			15800	112767	91

X Company borrows ₹ 1200000. at an interest rate of 14%.
The loan is to be repaid in 6 equal installment payable at the
end of 6 years. Prepare loan amortization schedule.

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Present value Annuity = $PMT \times PVIFA \% n.$

$1200000 = PMT \times 3.889$

$PMT = \frac{1200000}{3.889}$

$= 308563$

Year	Beginning Amt	Annual Installment	Interest	Per value Payment	Beginning of Borrowed Capital
	1200000	308563	3 = 14% x 1	4 = 2-3	
	1059437	308563	168000	140563	1059437
	899195	308563	148321	160242	899195
	716519	308563	125887	182676	716519
	508269	308563	100313	208250	508269
			71158	237405	

5 Suppose X Company has taken ₹ 2500000 has loan from a financial institution or bank with 11% P.a for 5 years find out EMI & Interest with the help of amortisation table.

$$\Rightarrow \text{Present value Annuity} = \text{PMT} \times \text{PVIFA } \% \cdot n$$

$$2500000 = \text{PMT} \times 3.696$$

$$\text{PMT} = \frac{2500000}{3.696}$$

$$= 676407$$

Year	Beginning ① Pmt	Instllment ②	Interest ③ = 11% X 1	P.V Payment ④ = 2-3	Beginning ⑤ = 1-4 Balance
1	2500000	676407	275000	401407	2098593
2	2098593	676407	230845	445562	1653031
3	1653031	676407	181833	494574	1158457
4	1158457	676407	127430	548977	609480
5	609480	676407	67043	609364	115

Compound Interest

21/9/23

1. An investor deposits Rs 100 in bank for 5 years at 8% interest? Find out the amount which he will in his account if interest is compounded:

1) Annually

2) Semi annually

3) Quarterly

$$F_n = P \left(1 + \frac{i}{m}\right)^n$$

$$F_{\text{ebb}} = P \times FVIF_{\frac{i}{m}, n \times m}$$

$$\text{Annual} = 100 \left(1 + \frac{0.08}{1}\right)^5$$

$$= 100 (1.4693)$$

$$= \underline{146.93}$$

$$\text{Semi Annually} = 100 \left(1 + \frac{0.08}{2}\right)^{5 \times 2}$$

$$= 100 \times 1.480$$

$$= 148$$

(or)

$$100 \times 1.48$$

$$\underline{148}$$

$$\text{Quarterly} = 100 \left(1 + \frac{0.08}{4}\right)^{5 \times 4}$$

$$= 100 \times 1.486$$

$$= \underline{148.6}$$

An Investor deposit ₹ 5000 in a bank for 6 years at 7% Interest. Find out the amount in which he will have in this account

1. Annually, 2. Semiannually & Quarterly

$$F_n = P \left(1 + \frac{i}{m}\right)^n$$

$$\text{Annually} = 5000 \left(1 + \frac{0.07}{1}\right)^6$$

$$= 5000 (1.5007)$$

$$= \underline{7503.65}$$

$$\text{Semi Annually} = 5000 \left(1 + \frac{0.07}{2}\right)^{6 \times 2}$$

$$= 5000 (1.5110)$$

$$= \underline{7555.34}$$

$$\text{Quarterly} = 5000 \left(1 + \frac{0.07}{4}\right)^{6 \times 4}$$

$$= 5000 (1.5164)$$

$$= \underline{7582.21}$$

- 3 Mr. makes a deposits of ₹ 20000 in Canada bank which takes 8% interest compounded annually for 8 years.

$$F_n = P \left(1 + \frac{i}{m}\right)^n$$

$$\begin{aligned} \text{Annually} &= 20000 \left(1 + \frac{0.08}{1}\right)^8 \\ &= 20000 (1.8509) \\ &= \underline{\underline{37018.60}} \end{aligned}$$

$$\begin{aligned} \text{Semiannually} &= 20000 \left(1 + \frac{0.08}{2}\right)^{8 \times 2} \\ &= 20000 (1.8729) \\ &= \underline{\underline{37459.62}} \end{aligned}$$

$$\begin{aligned} \text{Quarterly} &= 20000 \left(1 + \frac{0.08}{4}\right)^{8 \times 4} \\ &= 20000 (1.8845) \\ &= \underline{\underline{37690.81}} \end{aligned}$$

- 4 Suppose you deposit 1000 in a bank it takes 10% interest compounded annually. How much will the deposit grow after 8 years & 8 years?

$$F_n = P \left(1 + \frac{i}{m}\right)^n$$

$$\begin{aligned} \text{Annually} &= 1000 \left(1 + \frac{0.10}{1}\right)^8 \\ &= 1100 \end{aligned} \quad \begin{aligned} \text{Annually} &= 1000 \left(1 + \frac{0.1}{1}\right)^8 \\ &= 2143.58 \end{aligned}$$

$$\begin{aligned} \text{Semiannually} &= 1000 \left(1 + \frac{0.1}{2}\right)^{8 \times 2} \\ &= 1102.5 \end{aligned} \quad \begin{aligned} \text{Semiannually} &= 1000 \left(1 + \frac{0.1}{2}\right)^{8 \times 2} \\ &= 2182.87 \end{aligned}$$

$$\begin{aligned} \text{Quarterly} &= 1000 \left(1 + \frac{0.1}{4}\right)^{8 \times 4} \\ &= 1103.81 \end{aligned} \quad \begin{aligned} \text{Quarterly} &= 1000 \left(1 + \frac{0.1}{4}\right)^{8 \times 4} \\ &= 2203.75 \end{aligned}$$

5. Calculate the Compound value of ₹ 15000 at the end of 3 years at 12% Int Rate, when interest is calculated on the basis of yearly, half yearly & Quarterly.

$$F_n = P \left(1 + \frac{i}{m}\right)^n$$

$$\text{yearly} = 15000 \left(1 + \frac{0.12}{1}\right)^3$$

$$= 21073.92$$

$$\text{half yearly} = 15000 \left(1 + \frac{0.12}{2}\right)^{3 \times 2}$$

$$= 21277.78$$

$$\text{Quarterly} = 15000 \left(1 + \frac{0.12}{4}\right)^{3 \times 4}$$

$$= 21386.41$$

Mr Ganesh deposited at the end of each year ₹ 15000, ₹ 20000 & ₹ 25000, in his saving bank account. Interest rate of 6% you want to know its future value of deposits at the end of 5 years.

$$\text{yearly} = 15000 \left(1 + \frac{0.06}{1}\right)^5$$

$$= 20073.38$$

$$= 5000 \left(1 + \frac{0.06}{1}\right)^5$$

$$= 6691.12$$

$$\text{yearly} = 20000 \left(1 + \frac{0.06}{1}\right)^5$$

$$= 26764.51$$

$$= 10000 \left(1 + \frac{0.06}{1}\right)^5$$

$$= 13382.25$$

$$\text{yearly} = 25000 \left(1 + \frac{0.06}{1}\right)^5$$

$$= 33455.63$$